17 Adrian is going to make concrete.
He is going to use

180 kg of cement
375 kg of sand
1080 kg of stone

Cement, sand and stone are sold in bags.

1 bag cement	1 bag sand	1 bag stone
25 kg	22.5 kg	50 kg

Adrian already has

10 bags of cement
20 bags of sand
20 bags of stone

Work out what bags he needs to buy to make the concrete.

Cement already available:

	Bags	kg	
×10	1	25	×10
	10	250	

250 kg ≥ 180 kg
Enough cement.

Sand already available:

	Bags	kg	
×20	1	22.5	×20
	20	450	

450 kg ≥ 375 kg
Enough sand.

Stone already available:

	Bags	kg	
×20	1	50	×20
	20	1000	

1000 kg < 1080 kg

2 marks total
Work out all three available quantities

Stone shortfall = 1080 – 1000 = 80 kg.

80 ÷ 50 = 1.6 bags: buy 2 whole bags of stone.

1 mark
Round up: no extra cement or sand is needed

(Total for Question 17 is 3 marks)